

BEACONHOUSE NATIONAL UNIVERSITY



Assignment 02

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BNU ID: F2023-009

Total Marks: 10

Due Date: 9-11-2025

OS(SEC_C)

Course Instructor	Miss Noreen Khalid
Lab Instructor	Sir Ali Hafeez
Semester	Fall 2025

3 Nov 2025

Operating System

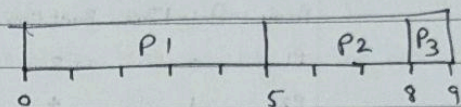
Assignment - 02

Monday

SAAD MUHAHAL

F2023-009

Question 1: Step 1: Gantt Chart

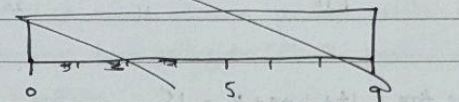


P1:

P2:

P3:

Time:



Non-Preemptive

(Finish time - Arrival time)

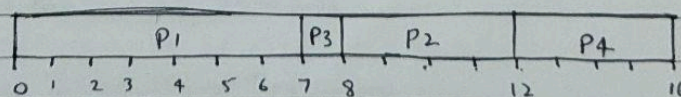
Process	Arrival time	Burst time	Start time	Finish time	Waiting Working Time	Turnaround time
P1	0	5	0	5	0	5
P2	2	3	5	8	3	6
P3	4	1	8	9	4	5

$$\text{Average Waiting Time} = \frac{(0+3+4)}{3} = \frac{7}{3} \approx 2.333$$

$$\text{Average Turnaround Time} = \frac{(5+6+5)}{3} = \frac{16}{3} \approx 5.333$$

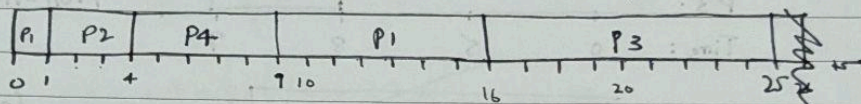
Question 2: SJF (Shortest Job First) - Non-Preemptive

Process	Arrival Time	Burst Time	Start Time	Finish Time	Waiting Time	Turnaround Time
P1	0	7	0	7	0	7
P2	2	4	8	12	6	10
P3	4	1	7	8	3	4
P4	5	4	12	16	7	11



Question 3: SJF (Shortest Remaining Time First) - Preemptive

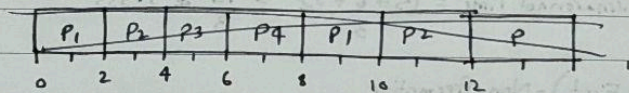
Process	Arrival Time	Burst time	Start Time	Finish Time	Waiting Time	Turnaround time
P1	0	8	0	16	0	16
P2	1	4	1	5	0	4
P3	2	9	16	25	14	23
P4	3	5	4	9	1	6



$$\text{Average waiting time} = \frac{14 + 1 + 0 + 0}{4} = \frac{15}{4}$$

Question 4: Round Robin Scheduling Time Quantum (Q) = 2 units

Process	Arrival time	Burst Time	Start time	Finish time	Waiting time	Turnaround time
P1	0	5	0	12	0	12
P2	1	4	2	6	1	5
P3	2	2	4	6	2	4
P4	3	1	6	7	3	4



$$\text{Average waiting time} = \frac{0 + 1 + 2 + 3}{4} = \frac{6}{4} = \frac{3}{2}$$

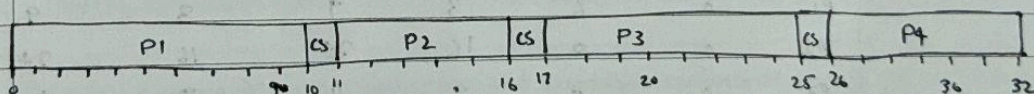
$$\text{Average turnaround time} = \frac{12 + 5 + 4 + 4}{4} = \frac{25}{4}$$

Question 5: Comparison & Reflection

1. Shortest Job first (SJF) algorithm gives the lowest avg. waiting time because it always selects the process with the smallest burst time first, reducing waiting time of all other processes.
2. The preemptive priority scheduling algorithm is best for real-time systems because high priority tasks can interrupt the process to ensure quick & predictable responses.
3. ~~the systems~~ Starvation occurs when a process waits indefinitely because other shorter or high priority processes keep getting the CPU. The problem often happens in SJF and priority scheduling Algorithms.
4. If time ~~quantum~~ quantum is too small, there will be many context switches, causing CPU overhead. If it is too large, the algorithm behaves like FCFS, leading to poor time response. Therefore it should be chosen near to avg. burst time for balance.

Question 6: Mixed Arrival with FCFS & Context Switching

Process	Arrival Time	Burst Time	Start time	Finish time	Waiting time	Turnaround time
P1	0	10	0	10	0	10
P2	3	5	11	16	8	13
P3	5	8	17	25	12	20
P4	9	6	26	32	17	23



CPU Utilization with Context switching Overhead

Total Burst time = $10 + 5 + 8 + 6 = 29$ units

Total Context Switching time = 3 switches $\times$ 1 unit = 3 units

Total time from start to end = 32 units

$$\text{CPU Util} = \frac{\text{Total Burst Time}}{\text{Total time}} \times 100$$

Without context switching

Process	Completion times	Turnaround time	Waiting time
P1	10	10	0
P2	15	12	7
P3	23	18	10
P4	29	20	14

$$\text{Average waiting time} = \frac{0+7+10+14}{4} = 7.75$$

$$\text{Average Turnaround time} = \frac{10+12+18+20}{4} = 15$$

With context switching

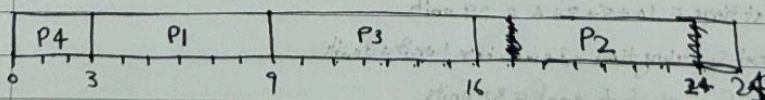
$$\text{Average waiting time} = \frac{0+8+12+17}{4} = 9.25$$

$$\text{Average Turnaround Time} = \frac{10+13+20+23}{4} = 16.5$$

Context switching increases average waiting time and average turnaround time due to overhead, and reduces CPU utilization.

Question 7: SJF (Preemptive) with Same Arrival Times

Process	Arrival time	Burst time	Start time	Finish time	Waiting time	turnaround time
P1	0	6	3	9	3	9
P2	0	8	16	24	16	24
P3	0	7	9	16	9	16
P4	0	3	0	3	0	3



$$\text{Average waiting time} = \frac{3+16+9+0}{4} = 7$$

$$\text{Average turnaround time} = \frac{9+24+16+3}{4} = 13$$

Comparison with SJF Non-preemptive

Since all processes arrive at the same time, SJF preemptive and SJF Non-preemptive yield identical results:

- Execution order: P4, P1, P3, P2
- Gantt chart, waiting time, and turnaround times are the same.
- Thus, there is no difference in average waiting time and average turnaround time.

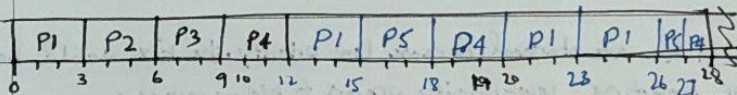
Effect of Preemption on Average Waiting time

Preemption in SJF (or SRTF) reduces the average waiting time when shorter jobs arrive after longer jobs have started, as it allows shorter jobs to be executed earlier. However, when all processes arrive simultaneously, there are no new arrivals to preempt the current process, so preemption has no effect. In this case, the average waiting time is the same for both preemptive and non-preemptive SJF.

Question 8:

Round Robin with Unequal Burst times : Quantum (Q) : 3 units

Process	Arrival time	Burst time	Start time	Finish time	Waiting time	Turnaround time
P1	0	9		26	17	26
P2	1	5		20	14	19
P3	2	3		9	4	7
P4	3	7		28	18	25
P5	4	4		27	19	23



$$\text{Average waiting time} = \frac{17+14+4+18+19}{5} \approx 14.4$$

$$\text{Average turnaround time} = \frac{26+19+7+25+23}{5} \approx 20.0$$

A smaller quantum improves responsiveness but increases context switching which results in increase in average waiting time and with larger quantum it generally decreases the time as less context switching.

The trade-off lies between quick response and efficiency.

Q9 Multi-level Queue Scheduling

Process	Queue	Arrival	Burst	Completion	Turnaround	Waiting
P ₁	Q ₁	0	5	8	8	3
P ₂	Q ₂	1	1	21	20	13
P ₃	Q ₂	2	3	6	4	1
P ₄	Q ₂	3	4	25	22	18
P ₅	Q ₁	4	6	14	10	4

P ₁ Q ₁	P ₂ Q ₁	P ₃ Q ₁	P ₄ Q ₁	P ₅ Q ₁	P ₂ Q ₂	P ₃ Q ₂	P ₄ Q ₂
0	3	6	7	8	11	14	

$$\text{Average } Q_1 \text{ WT} = 3 + 1 + 4 / 3 \approx 2.67$$

$$\text{Average } Q_1 \text{ TAT} = 8 + 4 + 10 / 3 \approx 7.33$$

$$\text{Average } Q_2 \text{ WT} = 13 + 18 / 2 = 15.5$$

$$\text{Average } Q_2 \text{ TAT} = 20 + 22 / 2 = 21.0$$

$$\text{CPU Utilization} = 25 / 25 \approx 100\%$$

$$\text{Throughput} = 5 / 25 = 0.2 \text{ processes/unit time}$$

Processes in Q₂ are starved because low priority. The fix would be to add aging or promoting low-priority process to higher queue.

Reflection Questions

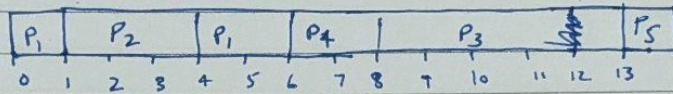
① Round Robin

this algorithm gives fair CPU sharing and good response time while keeping waiting time reasonable.

② Increase CPU overhead and idle time, which reduce CPU efficiency.

Q10 Priority Scheduling (Preemptive)

Process	Arrival	Burst	Priority	Completion	waiting	turnaround
P ₁	0	4	2	6	2	6
P ₂	1	3	1	4	0	3
P ₃	2	5	4	13	6	11
P ₄	3	2	3	8	3	5
P ₅	4	1	5	14	9	10



$$\text{Average waiting time} = \frac{2+0+6+3+9}{5} \approx 4.0$$

$$\text{Average turnaround time} = \frac{6+3+11+5+10}{5} \approx 7.0$$

When the higher-priority process arrives, CPU preempts the current one.

Non-preemptive version would delay P₂ until P₁ finishes.

Reflection Questions

- ③ SJF may starve long processes if short ones keep arriving. Aging gradually increases waiting process priority to prevent starvation.
- ④ AI can predict process behavior and dynamically adjust scheduling for better performance and fairness.